

November 2017  
Time : 2 hours

Marks available: 69  
Full marks:  $66\frac{2}{3}$

*No calculators may be used.*

1. Read the following theorem and the outline of its proof:

**Theorem.** Suppose that a complex-valued function  $f$  is analytic on an open disc  $D_R = \{z \in \mathbb{C} : |z - z_0| < R\}$ , where  $z_0 \in \mathbb{C}$  and  $R > 0$ . If  $z_0$  is a limit point of  $\{z \in D_R : f(z) = 0\}$ , then  $f(z) = 0$  for all  $z \in D_R$ .

*Proof.* If  $f(z)$  is not identically zero on  $D_R$ , then its Taylor series  $\sum_{k=0}^{\infty} a_k (z - z_0)^k$  must have at least one non-zero coefficient. Let  $n$  be the smallest integer such that  $a_n \neq 0$ . Since  $f(z_0) = 0$ , we know that  $n > 0$ , so

$$f(z) = (z - z_0)^n \sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k.$$

Put  $g(z) = \sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k$ . Then  $g(z_0) = a_n \neq 0$ , so  $g(z)$  will be non-zero on an open disc  $D_r = \{z \in \mathbb{C} : |z - z_0| < r\}$ , where  $0 < r < R$ . But then  $f(z) = (z - z_0)^n g(z) \neq 0$  for all  $z \neq z_0$  in  $D_r$ , a contradiction.

Now answer the following questions:

- (a) Define what is meant by  $f$  being *analytic* on the open disc  $D_R$ .
- (b) What is a *limit point* of a set of complex numbers?
- (c) How do we know that the Taylor series of  $f$  converges to  $f$  on the disc  $D_R$ ?
- (d) Why is  $f(z_0) = 0$ ?
- (e) Why does it follow from the fact that  $g(z_0) \neq 0$  that  $g(z) \neq 0$  for all  $z$  in a disc centred at  $z_0$ ?
- (f) What exactly is the contradiction referred to at the end of the proof?

[1, 3, 2, 2, 2, 2]

2. Define  $f: \mathbb{C} \rightarrow \mathbb{C}$  by  $f(z) = \begin{cases} \frac{z^5}{|z|^4} & \text{if } z \neq 0 \\ 0 & \text{if } z = 0 \end{cases}$ .

- (a) Let  $z = x + iy$ , where  $x, y \in \mathbb{R}$ . Use the binomial theorem to find the real and imaginary parts of  $f(z)$ .
- (b) Does  $f$  satisfy the Cauchy–Riemann equations at  $z = 0$ ? (A “Yes” or “No” answer is not sufficient.)
- (c) Is  $f$  differentiable at  $z = 0$ ? (Again, “Yes” or “No” is not enough.)

[3, 3, 2]

3. What is the radius of convergence of the power series  $\sum_{k=0}^{\infty} \frac{(k!)^2}{(2k)!} z^{2k}$ ? [5]

4. Calculate each of the following integrals using the illustrated path:

(a)  $\int_{\gamma} \bar{z} dz$

(b)  $\int_{\gamma} \frac{\tan z}{z^2 - 4} dz$

[9]

5. Evaluate  $\int_{\gamma} \frac{\operatorname{Re}(z)}{z - \frac{1}{2}} dz$ , where  $\gamma(t) = e^{it}$ ,  $0 \leq t \leq 2\pi$ . [8]

6. Let  $f$  be analytic on the open disc  $D_1 = \{z \in \mathbb{C} : |z| < 1\}$ . Show that if

$$|f(z)| \leq 1 - |z|$$

for all  $z \in D_1$ , then  $f(z) = 0$  on  $D_1$ . [5]

7. Use complex variable methods to evaluate

(a)  $\int_{-\infty}^{\infty} \frac{1}{x^2 + 2x + 5} dx$   
(b)  $\int_0^{\infty} \frac{x \sin x}{x^2 + 9} dx$

[7, 7]

8. Find a Laurent series expansion of  $f(z) = \frac{1}{z^2 - 3z + 2}$  that converges on the set  $\{z \in \mathbb{C} : 1 < |z| < 2\}$ . [3]

9. Show that for each  $a > e$ , the equation

$$e^z = az^n$$

has  $n$  solutions inside the unit circle. [5]